INTERNATIONAL ISLAMIC UNIVERSITY

ASSIGNMENT-01

STUDENT INFORMATION

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C241210
4th semester, 4FM

COURSE DETAILS

CSE-2427
Theory of Computation

LECTURER DETAILS

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SUBMISSION DATE

10/9/2025

REMARKS

FAHAD C241210

1)

(a) Define computability Theory.

A: Two states of an automaton are said to be compatible if there is no input string that can distinguish them.

(b) Constract DFA ~~from~~ the following language where alphabet is {0,1}

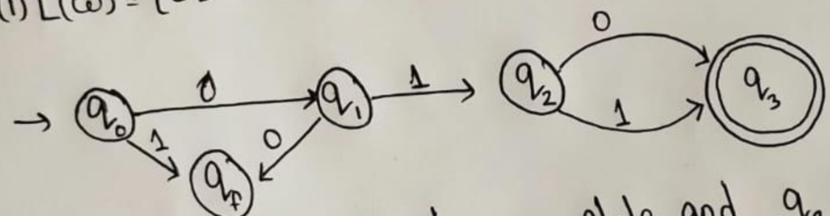
(i) {w|w begins with 01 and the length w is odd}

(ii) {w|w starts and ends with different symbols}

(iii) {w|w doesn't contain the substring 0011}

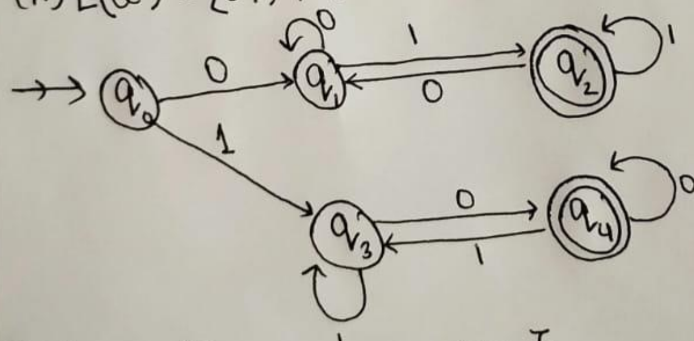
A: ~~(i)~~ Language alphabet is {0,1}

(i) L(w) = {010, 011, 01101, ... }

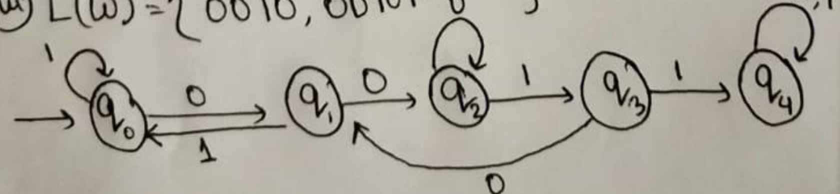


q_f rejected state, q_0 begining state and q_3 final state.

(ii) L(w) = {01, 10, 001, 110, 0101}



(iii) L(w) = {0010, 00101 ... }



(c) Write regular expressions for the languages described in 1(b).

A:

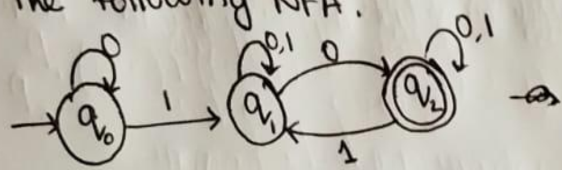
$\Rightarrow$ (i) $01((0|1)(0|1))*(0|1)$ → ~~starts and ends with different alphabet~~ starts with 01 and length odd.

(ii) $(0(0|1)*1) + 1(0|1)*0)$ → start and end alphabet different.

(iii) All binary strings that do not contain the substring 0011

$(0|1)* - (0|1)*0011(0|1)*$

2) (a) Show the steps of a computation of the string 011001 on the following NFA.



A: All the expression for NFA is,

$q_2 = q_1 0 + q_2 \cdot 1 + q_2 \cdot 0$ — (i)

$q_1 = q_2 \cdot 1 + q_1 \cdot 0 + q_1 \cdot 1 + q_0 \cdot 1$ — (ii)

$q_0 = \epsilon + q_0 \cdot 0$ — (iii)

From eq^n (iii) we get,

$q_0 = \epsilon + q_0 \cdot 0$

$= q_0 \cdot 0*$ [from arden's theorem]

From eq^n (i) we get,

$q_2 = q_1 \cdot 0 + q_2 \cdot 1 + q_2 \cdot 0$

$= q_1 \cdot 0 + q_2 \cdot (1+0)$

$= q_1 0 (1+0)*$ [from arden's theorem]

From eq^n (ii) we get,

$q_1 = q_2 \cdot 1 + q_1 \cdot 0 + q_1 \cdot 1 + q_0 \cdot 1$

Putting values of q_0 & q_2

3

$q_1 = (0^*)1 + q_1.0 + q_1.1 + (q_1.0(0+1)^*)1$

$= (0^*)1 + q_1((0+1) + 0(0+1)^*1)$

$= (0^*)1((0+1) + 0(0+1)^*1)^*$

Final Regular expression is.

$q_2 = (0^*)1((0+1)+0(0+1)^*1)^*0(0+1)^*$ (Ans)

(b) What are the languages described by the following regular expressions? write one sentence description for each ~~expre~~ language.

(i) $1^*(00)^*$

(ii) $(10 \cup 0)^*$

A:

(i) The language consist of all strings that starts with zero or more ones followed by zero or more pairs of zeros (00)

(ii) The language consists of all strings over {0,1} where every occurance of the symbol 1 must be immadiately followed by 0.

(c) Prove that every nondeterministic finite automaton has an equivalent deterministic finite automaton.

A: Let $N = (Q, \Sigma, \delta, q_0, F)$ be the NFA recognizing some language A. We construct a DFA $M = (Q', \Sigma, \delta', q_0', F')$ recognizing A. Before doing the full construction, let's first consider the easier case wherein N has no ε arrows. Later we take the ε arrows into account.

① $Q' = P(Q)$
Every state of M is a set of states of N. Recall the $P(Q)$ is the set of subsets of Q.

② For $R \in Q'$ and $a \in \Sigma$, let $\delta'(R, a) = \{q \in Q \mid q \in \delta(r, a)$ for some $r \in R\}$. If R is a state of M it is also a set of states of N, When M reads a symbol a in state R, it shows where a takes each state in R. Beacuse each state may go to a set of states, we take the union of all these ~~state~~ sets. Another way to write this expression is

$$\delta'(R, a) = \bigcup_{r \in R} \delta(r, a)$$

③ $q_0' = \{q_0\}$
M starts in the state corresponding to the collection containing just the start state of N.

④ $F' = \{R \in Q' \mid R$ contains an accept state of $N\}$
The machine M accepts if one of the possible states that N could be in at this point is an accept state.

Now we need to consider the ε arrows, To do so, we set up an extra bit of notation. For any state R of M, we define $E(R)$ to be the collection of states that can be reached from member of R by going only along ε arrows, including the members of R themselves. Formally for $R \subseteq Q$ let.

$E(R) = \{q \mid q$ can be reached from R by traveling along 0 or more ε arrows$\}$

Then we modify the transition function of M to place additional fingers on all states that can be reached by going along ε arrows after every step. Replacing $\delta(r,a)$ by $E(\delta(r,a))$ achives this effect. Thus

$$\delta'(R,a) = \{q \in Q \mid q \in E(\delta(r,a)) \text{ for some } r \in R\}.$$

Q) Prove that the class of regular languages is closed under the star operation.

A: let $N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1 construct $N = (Q, \Sigma, \delta, q_0, F)$ to recognize A_1^*

1 $Q = \{q_0\} \cup Q_1$
The states of N are the states of N_1 plus a new state.

2 The state q_0 is the new start state.

3 $F = \{q_0\} \cup F_1$
The accept states are the old accept states plus the new state

4 define δ so that for any $q \in Q$ and any $a \in \Sigma_\varepsilon$

$$\delta(q,a) \begin{cases} \delta_1(q,a) & q \in Q_1 \text{ and } q \notin F_1 \\ \delta_1(q,a) & q \in F_1 \text{ and } a \neq \varepsilon \\ \delta_1(q,a) \cup \{q_1\} & q \in F_1 \text{ and } a = \varepsilon \\ \{q_1\} & q = q_0 \text{ and } a = \varepsilon \\ \emptyset & q = q_0 \text{ and } a \neq \varepsilon \end{cases}$$

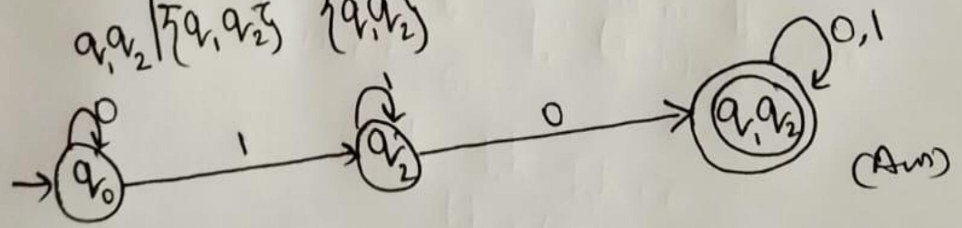
3) (a) Convert the following NFA to an equivalent DFA. Give only the portion of the DFA that is reachable from the start state.

A: transition table of NFA.

	0	1
q_0	$\{q_0\}$	$\{q_2\}$
q_1	$\{q_1\}$	$\{q_1, q_2\}$
q_2	$\{q_2, q_1\}$	$\{q_2\}$

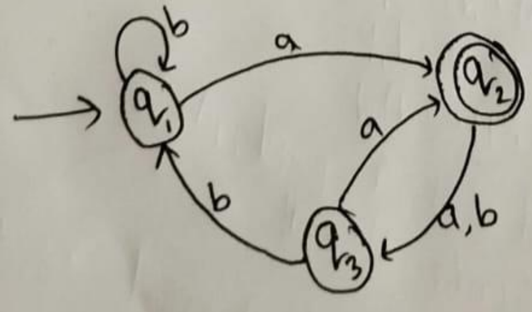
transition table of DFA

	0	1
q_0	$\{q_0\}$	$\{q_2\}$
q_2	$\{q_1 q_2\}$	$\{q_2\}$
q_1q_2	$\{q_1, q_2\}$	$\{q_1q_2\}$



(Ans)

(b) Convert the following DFA to a regular expression.



A: Equation from the given DFA.

$\cancel{q_2 = q_1 a + q_3 a}$ — (i) $\cancel{q_2 = q_1 a + q_3 a}$ — (ii)

$q_3 = q_2 a + q_2 b$ — (i)

$q_2 = q_1 a + q_3 a$ — (ii)

$q_1 = q_1 b + q_3 b + \epsilon$ — (iii)

From eqn (i),

$q_3 = q_2 a + q_2 b$

$= q_2(a+b)$

From eqn (ii),

$q_2 = q_1 a + q_3 a$

$= q_1 a + (q_2(a+b))a$ [Putting value of q_3]

$= q_1 a + q_2 (a+b)a$

useing arden's theorem.

$q_2 = q_1 a ((a+b)a)^*$

From eqn (iii),

$q_1 = q_1 b + q_3 b + \epsilon$

$= q_1 b + (q_2(a+b))b + \epsilon$

$= q_1 b + q_2 (a+b)b + \epsilon$

$= q_1 b + (q_1 a ((a+b)a)^*) (a+b) b + \epsilon$

$= q_1 b + q_1 a ((a+b)a)^* (a+b) b + \epsilon$

$= q_1 b + q_1 a ((a+b)a)^* (a+b) b + \epsilon$

$= q_1 (b + a((a+b)a)^* (a+b) b) + \epsilon$

Applying Arden's theorem,

$= (b + a((a+b)a)^* (a+b) b)^*$

as we know final state is q_2 and,

$q_2 = q_1 a ((a+b)a)^*$

So the Regex $= (b + a((a+b)a)^* (a+b) b)^* a ((a+b)a)^*$

(Ans)